

A Unified Evaluation Framework for Privacy-Preserving Data Publication Techniques

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Abstract

Privacy-preserving data publication (PPDP) aims to reduce privacy risks before releasing datasets. As alternative techniques, (i) *data de-identification* alters original data values through generalization or suppression to weaken their link to data subject identities, whereas (ii) *synthetic data generation* produces artificial datasets that do not contain any original data yet preserve statistical similarity to the original data.

Because their outcomes lack direct comparability, it remains unclear which technique provides better privacy protection while retaining data utility. To address this gap, we present: (i) a novel quantitative evaluation framework for comparing privacy and utility across synthetic and de-identified datasets, and (ii) a comparative study based on a location optimisation task and a predictive machine learning application.

Our results reveal a nuanced trade-off. De-identified datasets perform better on traditional statistical privacy metrics but are more vulnerable to adversarial attacks such as attribute inference. Conversely, utility outcomes vary widely by use case, indicating that the selection of a PPDP technique remains highly dependent on the application context.

Keywords: data privacy, privacy-preserving data publishing, synthetic data, data de-identification, generative model, anonymized data

1 Introduction

Privacy concerns and regulations such as the GDPR in Europe, HIPAA in the United States, and POPIA in South Africa can restrict the sharing of sensitive data, despite the importance of such data for scientific and technological advancement. To address this challenge, privacy-enhancing technologies such as *Federated Learning* and *Secure Multi-party Computation* reduce the need for direct data sharing through decentralised computation. However, these approaches rely on pre-specified computations and lack the flexibility and robustness of direct data access. Consequently, techniques for privacy-preserving data publication (PPDP) remain essential for enabling practical and accountable data sharing.

There are two main PPDP strategies for tabular and structured data: (i) de-identification via generalising and suppressing personally identifiable information; or (ii) synthetic data generation via multivariate distribution modelling and sampling. While the former method is well-studied [3, 15, 38], it requires expertise and data-domain transformations, and still poses residual privacy risks upon data release. While the latter method breaks direct links to the original data and provides “first glance” similarity, its production is strongly dependent on generative algorithms and is costly, especially for high-dimensional data.

The relative effectiveness of these strategies remains actively debated in the literature. Cynthia Dwork, often regarded as the godmother of *Differential Privacy*, argues against data de-identification [8], suggesting that this strategy fails to achieve a satisfactory privacy–utility trade-off. In contrast, some researchers describe synthetic data as the ‘*holy grail of data sharing*’ [9]. However, other scholars argue that synthetic data is not necessarily a superior alternative [36, 37], as its performance is not deterministic and the resulting privacy gains are often insufficient to justify the computational overhead.

Resolving this debate requires a rigorous comparative evaluation of the privacy and utility outcomes of both PPDP techniques, which is lacking in the literature. This is in part because the evaluation practices commonly used for PPDP techniques differ substantially, making their privacy results difficult to compare directly. For synthetic data, holdout-based assessment [29] is widely used to estimate privacy and utility by comparing released data with unseen original records. The holdout set serves as a useful, principled reference point for both privacy and utility, enabling more consistent comparisons across different PPDP approaches. On the other hand, evaluations of de-identified data typically focus on quasi-identifiers (QIDs), reflecting their central role in re-identification risk. However, holdout-based evaluation is rarely applied to de-identified data [21, 26], while QID-specific analyses are not considered when

evaluating synthetic data [36, 37].

We present a common task framework that incorporates both evaluation practices. By combining holdout-based assessment with QID-aware analysis under consistent threat models, our framework enables the principled, comparative evaluation between data de-identification and synthetic data release.

We evaluate the framework through a controlled comparative study involving two complementary and realistic PPDP applications: (i) processing patient and location data (the *Patients* dataset) to determine the optimal placement of pharmacy outlets [41]; (ii) performing regression and classification tasks to predict individuals’ annual income based on attributes such as occupation and education (the *ACSIIncome* dataset [7]). Our results show that de-identified datasets appear more privacy-preserving under traditional statistical metrics, yet are more susceptible to attribute inference attacks than synthetic data. In terms of utility, we demonstrate that synthetic data and de-identified data achieve optimal utility in different use cases. While these insights aid practitioners in better understanding trade-offs, reaching a definitive conclusion requires a greater variety of settings and a joint effort from the research community.

Contribution/novelty statement. Our main contributions are twofold:

1. We establish a novel framework for the systematic and comparative evaluation of both PPDP approaches. The framework integrates relevant privacy and utility metrics, ensures applicability to both synthetic and de-identified datasets, and combines QID-based assessment with holdout-based evaluation. As secondary contributions, we conduct a qualitative analysis of metric applicability (presented in Table 1) and develop a distance formulation grounded in prior research [14, 41]. The framework implementation is open-sourced [1].
2. We report the main results of a comparative study across two relevant application cases, demonstrating that the choice of a PPDP strategy depends strongly on the application context and the acceptable level of privacy risk.

The remainder of this paper is structured as follows. Section 2 discusses the background. Then, Section 3 introduces the evaluation framework. Section 4 then discusses the experimental setup, and Section 5 presents the main results. Section 6 provides discussion, and Section 7 discusses related work. Finally, Section 8 concludes this paper.

2 Background

2.1 Privacy-Preserving Data Publication (PPDP)

Data publication poses significant privacy risks, especially when conducted with consolidated, large datasets involving many data subjects. We articulate the privacy threats relevant to this context using the LINDDUN privacy threat modelling framework [23]. The undue sharing of raw datasets is often considered *excessive* or *unnecessary* (*Data disclosure* threats in LINDDUN), and may lead to loss of plausible deniability (*Non-repudiation*), of user control in the overall proliferation of personal data (*Unintervenability*), and the overall sharing may be considered unlawful (*Non-compliance*). Furthermore, the shared data may be used to infer the data subject’s involvement (*Detecting*), to infer specific attributes (*Linking*), or to learn the data subject’s identity (*Identification*). PPDP refers to the deliberate use of proactive measures —such as de-identification¹ and synthetic data generation— to mitigate or reduce such privacy risks.

Data de-identification. Data de-identification protects individuals’ private information through targeted modification of the dataset. This is achieved using techniques such as generalisation, suppression, swapping, noise addition, and bucketization, which are applied to meet specific privacy criteria, such as k -anonymity [35]. Various tools, such as ARX [31], and sdcMicro [40] implement these techniques. User-defined hierarchies are used to support data generalisation. An optimal combination of generalisation and suppression for each attribute is typically found using heuristic search methods [6].

Synthetic data generation. The generation of tabular synthetic data follows a two-step process: (i) a statistical model or machine learning (ML) model (such as a *Generative Adversarial Network* [42], or *Diffusion model* [18]) is trained to approximate the properties of the original data; and (ii) a synthetic dataset is sampled from the resulting model. Privacy is achieved by providing a computer-generated surrogate representation of real data records. Existing libraries and tools, such as Synthcity [32] and SDV [28], provide these capabilities.

2.2 Privacy and Utility Evaluation

Ensuring the privacy and utility is a prerequisite for PPDP [24]. Privacy involves protecting sensitive information by actively preventing re-identification and other disclosures. Moreover, utility refers to the ability of PPDP data to remain useful for its intended applications. Owing to a fundamental incompatibility between both types of outcomes, the state of the art in de-identification and synthetic data generation uses different privacy and utility metrics.

Evaluating de-identified data. Privacy metrics are used to define the criteria for creating de-identified data. Most notably, k -anonymity [35] is a widely used metric. It states that each record is indistinguishable from at least $k - 1$

¹In literature, ‘de-identified data’ is also referred to as ‘*anonymised data*’ [37]. In this paper, we consistently use ‘de-identification’ to refer to a broader class of techniques that reduce the risk of identifiability through generalisation and suppression.

other records based on quasi-identifiers (QIDs) (e.g., attributes such as *age* or *gender*), that, when combined, could uniquely identify an individual. These groups of indistinguishable records, known as equivalence classes, are often obtained by combining techniques such as generalisation, suppression, swapping, noise addition, and bucketisation. Over time, l -diversity, t -closeness, and δ -disclosure, as extensions of k -anonymity, have been proposed to enhance privacy protection.

On the other hand, widely recognized utility metrics such as *Precision* [39], *Loss* [16], *Non-uniform entropy* [30], and *Average class size* [20] are often used to derive a utility score that is independent of specific use cases. Despite their popularity, these theoretical metrics have two significant limitations. First, they often do not correlate directly with practical, use-case-specific utility. Second, these metrics assume that data can be partitioned into equivalence classes, or that generalisations have been applied using hierarchy-based methods. These assumptions fail *under realistic threat models*, in which adversaries may exploit information irrespective of data partitions, and *under practical utility settings*, where practitioners use the data as provided for specific downstream tasks.

Evaluating Synthetic data. Unlike de-identified data, the level of privacy protection for synthetic data cannot be estimated upfront, due to the stochastic nature of the training and sampling processes in the generative model. The generative model learns to approximate the distribution of real data using objective functions based on the similarity between the approximated and real distributions. While similarity-based metrics, such as *Nearest Neighbor Distance* [25], *Closest Value Ratio* [28], *Similarity Ratio* [44], and *Nearest Neighbor Accuracy* [19], can be considered privacy metrics, research shows that the results of similarity tests do not indicate the model’s robustness to inference attacks. Hence, attack-based evaluation, such as *Attribute Inference Attack* (AIA) [11] and *Membership Inference Accuracy* (MIA) [37], may be better suited for estimating obscurity of sensitive properties [27].

Utility metrics for synthetic data are often scenario-based and depend on the specific task [12,13]. The usefulness of synthetic data is measured by how well it retains essential properties, typically quantified as the performance loss relative to using the original datasets. This is evaluated in terms of the overall suitability for downstream applications, such as machine learning tasks [5].

3 Evaluation Framework and Metrics

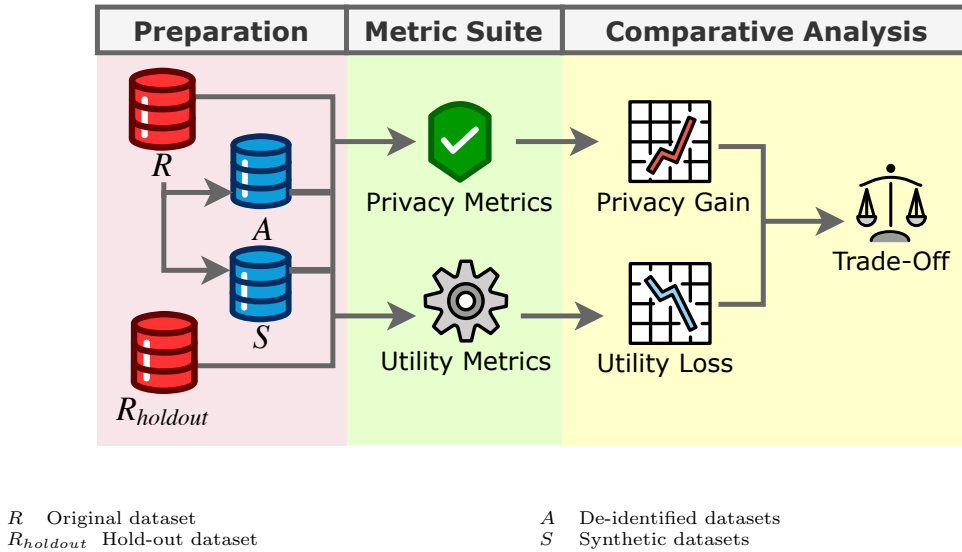


Figure 1: Overall evaluation framework.

Figure 1 depicts the unified evaluation framework that addresses the following key challenges: (i) **Establishing unified privacy and utility evaluation protocols:** we define protocols for consistently comparing the *privacy* and *utility* of synthetic and de-identified datasets (Section 3.1); (ii) **Establishing a standard suite of privacy metrics:** privacy metrics used in literature for synthetic data and de-identified data differ. We therefore survey privacy metrics, assess their applicability to both synthetic and de-identified data (Section 3.2), and adapt existing metrics where possible (Section 3.3). The resulting suite covers *similarity-based*, *model-based*, and *attack-based* privacy metrics. (iii) **Defining downstream task-based utility metrics:** finally, we introduce utility metrics grounded in downstream analytical tasks ², as detailed in Section 3.4.

²The privacy metric suite is comprehensive and exhaustive; however, we argue that utility metrics should depend on the specific application. Our utility evaluation protocol is designed to be metric-agnostic and can be extended to any new utility metric.

3.1 Evaluation Protocols

3.1.1 Privacy evaluation protocol.

We extended a hold-out-based assessment [29] that uses the original dataset (R) and a complementary hold-out set ($R_{holdout}$) drawn from the exact same statistical distribution as R . R is subjected to active privacy preservation, either by performing de-identification (leading to variant A), or synthetic data generation (leading to S). Hence, privacy evaluation is conducted using three types of datasets: (i) R , which represents the worst-case privacy baseline; (ii) $R_{holdout}$, which serves as an achievable privacy reference; and (iii) $M \in \{S, A\}$ which is a privacy-enhanced dataset obtained by applying a PPDP technique.

De-identification modifies only the quasi-identifiers (QIDs) while leaving other attributes unchanged. To enable a more nuanced comparison of the privacy guarantees associated with QIDs, we evaluate privacy along two attribute dimensions: (i) *QID-columns*: privacy metrics are computed using only QIDs. (ii) *All-columns*: privacy metrics are computed using all available attributes, representing a stronger privacy evaluation.

Privacy gain. Assuming that all privacy metrics are defined such that *higher values correspond to better privacy*, let $\mathcal{P}_d(\cdot)$ denote a privacy metric evaluated under the dimension $d \in \{qid, all\}$, we define the *privacy gain* $\mathcal{PG}_d$ of $M \in \{S, A\}$ under $\mathcal{P}$ and d as:

$$\mathcal{PG}_{\mathcal{P},d}(M) = \frac{\mathcal{P}_d(M) - \mathcal{P}_d(R)}{\mathcal{P}_d(R_{holdout}) - \mathcal{P}_d(R)} \quad (1)$$

This formulation anchors privacy measurements between R and $R_{holdout}$. Hence, the resulting privacy gain can be interpreted as:

- $\mathcal{PG}(M) < 0$: privacy is worse than the original dataset.
- $\mathcal{PG}(M) = 0$: privacy is on par with the original dataset.
- $\mathcal{PG}(M) = 1$: privacy matches that of the holdout dataset.
- $\mathcal{PG}(M) > 1$: privacy exceeds the holdout baseline (preferable).

Because the same original and holdout datasets are used as anchors, privacy gains are directly comparable across PPDP techniques in different dimensions.

3.1.2 Utility evaluation protocol.

To evaluate downstream utility, we adopt a fixed protocol across all datasets. For each released dataset variant $M \in \{S, A\}$, we apply a utility metric derived from a downstream utility task. As a reference baseline, we apply the same utility metric to R . This baseline represents the preferred utility under the chosen downstream utility task.

Utility loss. Assuming all utility metrics are defined such that *higher values correspond to better utility*, let $\mathcal{U}(\cdot)$ denote a utility metric. *Utility loss* $\mathcal{UL}$ of $M \in \{S, A\}$ is then defined as:

$$\mathcal{UL}(M) = \mathcal{U}(R) - \mathcal{U}(M). \quad (2)$$

A positive $\mathcal{UL}(M)$ value indicates a degradation in utility relative to the original-data baseline, while $\mathcal{UL}(M) = 0$ indicates no utility loss. Because all methods are evaluated on the same holdout set, $\mathcal{UL}(M)$ provides a directly comparable measure of utility degradation across PPDP techniques.

3.2 Privacy Metrics

We argue that privacy evaluation should not rely on arbitrarily selected metrics without proper justification or quality assessment. Therefore, we conduct a comprehensive qualitative analysis of the feasibility and relevance of privacy metrics for both PPDP techniques. This analysis is grounded on an explicit attacker model, which informs the selection and interpretation of the privacy metrics.

Privacy threat model or attacker model. We assume that an adversary has unrestricted access to the complete released dataset (A or S), and we evaluate two complementary attacker assumptions. Under the **maximum-knowledge intruder model** [34], the attacker can apply a wide range of transformations and inference techniques in an attempt to reverse the privacy-preserving mechanism, and evaluate the quality of each outcome with R (e.g., in terms of likelihood or similarity to real data)³. Under the **public knowledge model** [29], the attacker is assumed to have access to auxiliary data sampled from the same underlying distribution as the original dataset R , which is represented as $R_{holdout}$. A practical example of such public knowledge is census data, where newly released datasets may coexist with historical releases sharing the same attributes.

Selected privacy metrics. The results of metric selection are listed in Table 1. For each metric, we assess its applicability and portability to de-identified data and make a grounded selection for inclusion in our evaluation framework. The final column of Table 1 provides explicit rationale for inclusion or exclusion. Privacy metrics

³This represents a worst-case assumption and may be unrealistic in practice.

commonly used to evaluate de-identified data, such as k -anonymity, l -diversity, and t -closeness, essentially quantify the level of ambiguity within the released data set. However, in the case of synthetic data, such ambiguity is inherent, as all released data records are artificially generated without any direct, one-on-one link to records in the original data set. For this reason, we exclude such metrics from our study. To ensure diversity, we include applicable metrics of each class: (i) *similarity-based*, (ii) *model-based*, and (iii) *attack-based* or adversarial privacy metrics.

To ensure consistent interpretation across privacy metrics, we harmonise their directions (shown in the third column of Table 1) such that higher values always indicate stronger privacy protection.

3.3 Privacy metric adaptations.

In de-identification, (i) categorical attributes *QID columns* are generalized to higher-level concepts (e.g., ‘Six-Grade’ becomes ‘Primary School’), and (ii) numerical *QID columns* replace exact values with ranges (e.g., 25 becomes [20 – 30]). In consequence, the QIDs in A have different domains from those in S and R , which hinders direct comparability. To address this, we apply two types of adaptations, which are both based on established methods for distance computation and transformation in mixed-domain data using pre-defined hierarchies [14, 41]:

Adaptation 1: Distance calculation. Similarity-based metrics such as NND, CVR, SR, NNA, and AIA-C require distance computations between records. Distance hierarchies represent categorical and numerical values within a hierarchical structure and measure similarity as the path length between nodes. For numerical attributes, distances are computed using the L_1 distance⁴ in the original domain. Identical generalized ranges yield a distance of 0, whereas mismatched ranges are assigned the maximum distance. When comparing an original value with a generalized range, a normalized thresholded distance is used: values within the range are computed using L_1 distances among the possible values in the generalized domain, while values outside the range receive the maximum distance⁵. For categorical attributes, distances are computed using the generalization hierarchy, where the distance equals the number of edges between nodes. Matching generalizations yield a distance of 0, whereas mismatched generalizations receive the maximum number of edges. Attribute-level distances are normalized and aggregated to obtain a record-level distance.

Our distance evaluation does not favor any PPDP approach, because distance hierarchies provide a unified representation in which original and generalized values are comparable within the same distance space. Hence, distances between records from R , S , and A are computed under identical assumptions.

Adaptation 2: Domain transformation. Model-based metrics such as DR, SS, AIA-R, and MIA require datasets to share the same domain. To achieve this, generalized QID values in A are mapped back to the original domain through random sampling [41]. Each generalized value is replaced with a value uniformly sampled from the original values it represents (e.g., range [1 – 5] is replaced with a random value from {1, 2, 3, 4, 5}). This yields a plausible realization of the original data that is consistent with the generalization, without assuming a specific distribution.

3.4 Utility Metrics

Statistical utility metrics can be applied to evaluate the utility of any PPDP method. We use random count queries, as they provide a clearer sense of practical utility than summary statistics. Because downstream task utility is typically scenario-dependent, we evaluate predictive and optimisation tasks as representative downstream utilities.

Statistical Utility (Random Count Queries). Statistical utility is assessed using random count queries, a widely adopted approach for evaluating how well low-order statistics and attribute correlations are preserved under privacy transformations [22, 33]. Each query computes aggregate statistics of the form

COUNT(*) WHERE x IN R_x AND y IN R_y AND

on A or S , and the results are compared against those obtained from the original dataset R . The resulting *Count Query Error* (CQE) quantifies deviations in statistical correlations induced by the privacy mechanism. For generalised attributes, we assume that original values are uniformly distributed within each generalisation interval, allowing estimated counts to be derived. CQE directly reflects statistical utility loss.

Downstream Task Utility. Utility is evaluated by assessing how well the privacy-enhanced datasets support downstream analytical tasks. We consider two representative settings: (i) **Machine learning analytics.** A common application of PPDP is enabling machine learning (ML) analysis. Predictive performance is evaluated by training an ML model on A or S and measuring its test accuracy on $R_{holdout}$. The resulting accuracy, referred to as *ML Efficacy* (MLE), quantifies how well predictive performance is preserved. When attributes are generalised, the same generalisation procedure is applied to $R_{holdout}$ to ensure that the training and testing data reside in the same attribute domain [41]. (ii) **Optimisation tasks.** Optimisation utility is measured using the Resource Allocation Utility (RAU) metric [6]. An optimisation solution is derived from A or S and applied to the original dataset R . The resulting score is then normalised against the score obtained when optimising directly on R . Because RAU directly captures optimisation-related utility loss, we report RAU without further transformation. Higher RAU values indicate greater performance degradation and therefore lower utility.

⁴Our hierarchical and threshold-based distance calculation can be readily extended to other distance norms.

⁵Assigning the maximum distance to dissimilar values does not affect the final computation; rather, it acts as an effective filter because the privacy metrics consider only the nearest neighbours.

Table 1: Overview of the privacy metrics considered for this study, with motivations for including or excluding them.

Name	Abbrv.	↑	Description	Motivation
SIMILARITY- / DISTANCE-BASED PRIVACY METRICS				
Nearest Neighbor Distance	NND	↑	The mean distance of the released records to their closest records in the original dataset [25, 29, 32].	*** Similarity-based metrics remain relevant because they quantify how closely released data resembles the original data, providing a proxy for the amount of information preserved.
Close Value Ratio	CVR	↓	The percentage of released records that are similar to records in the original dataset (with a fixed threshold) [32].	*** Higher similarity indicates that more information about the original data is retained, which may increase the risk of privacy leakage.
Similarity Ratio	SR	↓	The percentage of released records that are similar to records in the original dataset (where the statistics of the original dataset define the threshold) [32, 44].	***
Nearest Neighbor Accuracy	NNA	↑	The sum of the dissimilarity percentages between the released dataset and the original dataset, and the released dataset to itself [43].	***
Hitting Ratio	HR	↓	The percentage of identical records between the released dataset and the original dataset [25, 28, 32].	🗑 De-identified data are real data with modified QIDs. This disadvantages de-identified data and skews results.
MODEL- / DISTRIBUTION BASED PRIVACY METRICS				
Detection Rate	DR	↑	The classification accuracy measures how accurately a classifier can distinguish records in the original dataset from those in a released dataset [28, 32].	🔧 A higher detection rate indicates greater deviation from the original dataset's distribution and, thus, potentially stronger privacy protection.
Structural Similarity	SS	↓	The similarity of the multivariate correlations between the released dataset and the original dataset. The multivariate distribution is derived through both structure learning and parameter learning of a Bayesian network.	🔧 Measuring how well the released data preserves the multivariate distribution of the original data can provide insights into its privacy strength.
Data Likelihood	DL	↓	The fidelity and similarity of the privacy-enhanced records by fitting the privacy-enhanced dataset to the distribution of the original dataset [28].	🔧 Measuring how much the distribution of the released dataset deviates from the original dataset helps quantify privacy.
ATTACK-BASED / ADVERSARIAL PRIVACY METRICS				
Attribute Inference Attack:			The accuracy or error of an active inference attack to derive or deduce sensitive values in the original dataset [13, 28, 32, 36, 37].	The extent to which a released dataset limits sensitive attribute inference reflects its privacy protection. AIA-based evaluation provides a model-agnostic framework applicable to both PPDP approaches.
- Accuracy	AIA-C	↓	- the target sensitive value is categorical.	***
- Regression error	AIA-R	↑	- the target sensitive value is numerical.	🔧
Membership Inference Attack	MIA	↓	The accuracy of successfully deducing whether a particular record is contained in the original dataset used to generate the released data [11, 27, 32, 34, 36, 37].	🗑 In de-identification, membership is a given: without record suppression, any original record will also appear in the de-identified dataset.
Reconstruction Attack	RA	↓	The error and accuracy of an adversary's reverse-engineered data records using synthetic data and auxiliary information [17].	🗑 Different variants rely on different threat assumptions [4, 10]. Finding a single assumption that applies to both cases is difficult, so they are incomparable.
Singling Out	SO	↓	The accuracy of uniquely identifying records through unique combinations of column values [11].	🗑 Under k -anonymity, the success rate is bounded by $\frac{1}{k}$, whereas in synthetic data, plausible deniability depends on the specific generative process. These differing assumptions make singling out comparisons across data types difficult to interpret and incompatible.
Record Linkage	RL	↓	The accuracy of establishing linkage to an external public dataset (that does not share the same distribution as the original dataset) through the released dataset [11, 25].	🗑 Assuming the availability of different external public datasets may lead to various outcomes.
*** Adaptation 1: Customized Distance calculation 🔧 Adaptation 2: Domain transformation 🗑 Excluded ↑ higher values indicate stronger privacy protection ↓ lower values indicate stronger privacy protection				

4 Experimental Setup

We perform an experimental validation of the established evaluation framework over two different use case. These experiments are conducted on a system equipped with an AMD Ryzen 9 5950X CPU and 128 GB of memory. This section provides a detailed description of the experimental setup.

4.1 Experiment datasets

We selected two datasets: (i) the *Patients* dataset and (ii) the *ACSIncome* dataset. The *Patients* dataset has previously been used to study the effects of de-identification in an optimisation use case [6]. It contains heart disease health records and demographic information (e.g., gender, age, income, and zip code) for 33,268 U.S. citizens. The 2018 *ACSIncome* dataset, extracted using Folktables [7], contains attributes similar to those of the widely used Adult dataset [2] while offering greater customizability. This dataset contains 1,664,500 records with 10 attributes and is suitable for training ML models to predict individuals’ income. Table 2 summarises both datasets, highlighting quasi-identifiers (QIDs) with underlining and sensitive attributes with dotted underlining⁶.

Table 2: Dataset descriptions of *Patients* and *ACSIncome*.

Dataset	Size (n)	Categorical	Numerical
<i>Patients</i>	33 268	<u>gender</u> , <u>stroke</u> , <u>ZIP</u> , <u>hypertension</u> , <u>heart_disease</u>	<u>smoking_status</u> , <u>age</u> , <u>bmi</u> , <u>Income</u> , <u>glucose_level</u>
<i>ACSIncome</i>	1 664 500	<u>COW</u> , <u>SCHL</u> , <u>MAR</u> , <u>OCCEP</u> , <u>POBP</u> , <u>SEX</u> , <u>RAC1P</u>	<u>AGEP</u> , <u>WKHP</u> , <u>PINCP</u>
		<u>underline</u> Quasi-identifier (QID)	<u>dotted underline</u> Sensitive attribute

4.2 Preparation and Preprocessing

Dataset preparation. We first preprocess the *Patients* dataset by imputing missing data. As a result, 32.9% of the rows are modified to impute values for the *smoking_status* and/or *bmi* attributes. The input dataset (i.e., *ACSIncome* or *Patients*) is then split into (i) a data set R containing 80% of the data and (ii) a hold-out set $R_{holdout}$ containing the remaining 20%. As shown in Figure 1, R is used to create S or A , while $R_{holdout}$ is utilised for validation.

Generation of privacy-enhanced datasets. Each de-identified dataset A is created using the ARX [31] library. Within this library, three parameters must be specified: (i) QIDs, as highlighted in Table 2; (ii) generalization hierarchies, which are adopted from [6] for the *Patients* and [41] for the *ACSIncome* dataset; and (iii) the k -anonymity privacy model, with k set to 5, 10, and 20. ARX provides utility heuristics to help practitioners select the optimal de-identified dataset from the available outputs. However, since we aim to evaluate privacy and utility using our chosen metrics, we assess all available de-identified datasets to ensure a fair comparison that is not influenced by such heuristics. These settings lead to 502 unique de-identified versions of the *Patients* dataset and 69 470 unique de-identified versions of the *ACSIncome* dataset.

Synthetic datasets S are generated using the SDV [28] and Synthcity [32] libraries. Three distinct generative models are applied: (i) CTGAN [42], (ii) TVAE [42], and (iii) TabDDPM [18]. The model parameters are set to ensure that model training is completed within a reasonable time. Specifically, the CTGAN and TVAE models from the SDV library are used with default parameters, except that the batch size was set to 1 000. Similarly, the TabDDPM model is used with default settings, except that the number of hidden layers is increased to 8 and the dataset-specific targets are set to *heart_disease* for *Patients* and *PINCP* for *ACSIncome*. To avoid uncertainty and sampling bias, five synthetic datasets, each matching the size of R , are generated for each trained model, resulting in a total of 15 synthetic datasets.

4.3 Performing Evaluation

Privacy evaluation. We calculate all privacy metrics selected in Table 1 by applying the protocols described in Section 3.1.1. The AIA targets are restricted to the sensitive attributes within each dataset, i.e., those highlighted as such in Table 2. The AIA metric is implemented with k -nearest neighbours and XGBoost.

Utility evaluation. We evaluate utility using the protocol described in Section 3.1.2. The *Count Query Error* (CQE) metric (Section 3.4) is dataset-agnostic and applied to both *ACSIncome* and *Patients*.

In addition, each dataset is evaluated using a task-specific utility metric aligned with its application. For *ACSIncome*, ML models predict individual income (*PINCP*) using (i) a binary classifier distinguishing incomes above or below \$50,000 and (ii) a multiclass classifier predicting the ranges [0–20,000[, [20,000–100,000[, and > 100,000. Both models use XGBoost, with categorical attributes one-hot encoded and numerical features standardised. To train on A and evaluate on $R_{holdout}$, we align the domains by applying the same generalisation to $R_{holdout}$; numerical ranges

⁶In the *ACSIncome* dataset, only the *PINCP* representing an individual’s exact income is labeled as a sensitive attribute. For *Patients*, these include medical attributes (*hypertension*, *heart_disease*, *avg_glucose_level*, *bmi*, and *stroke*) and *Income*.

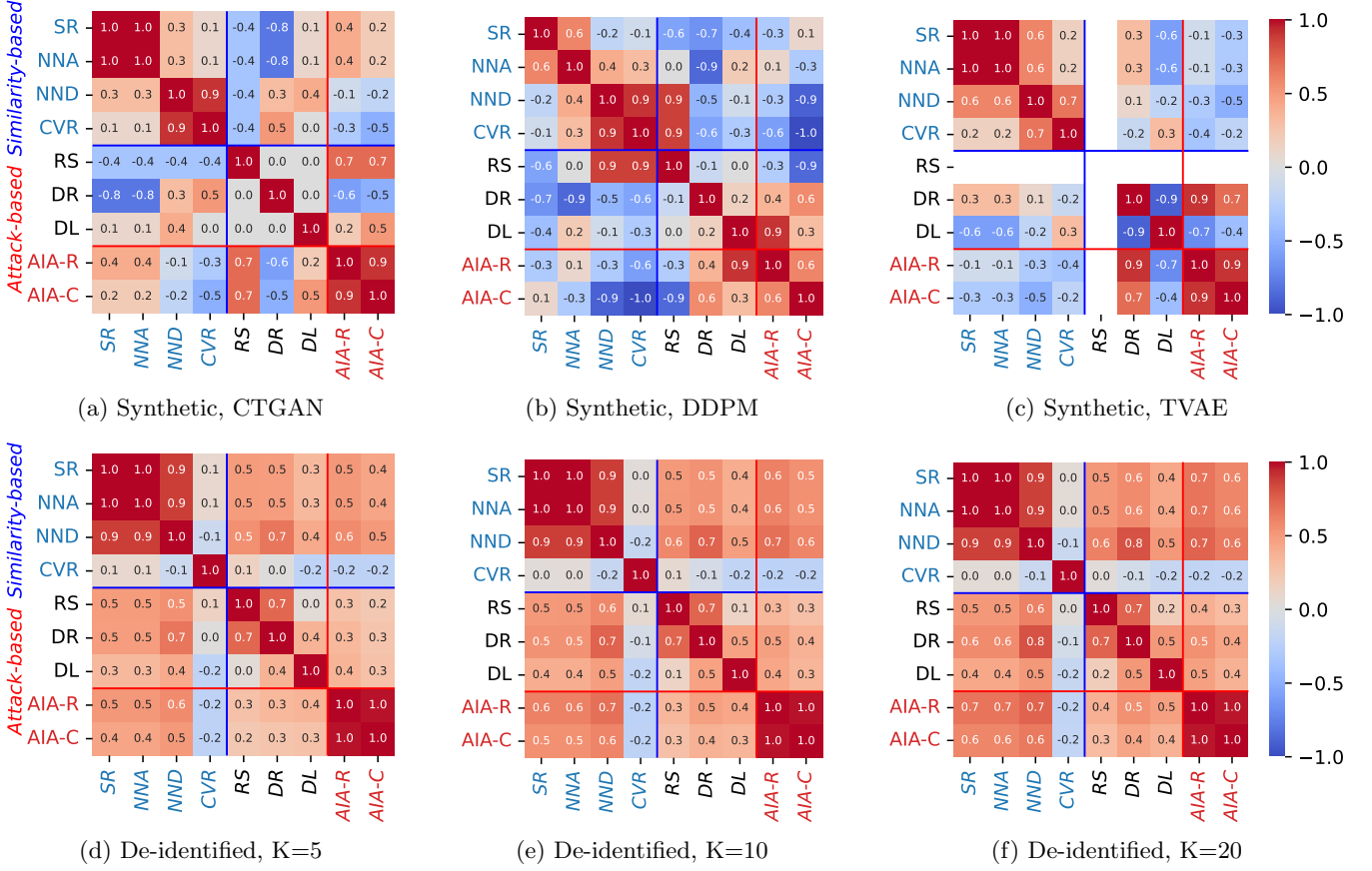


Figure 2: Spearman correlation heatmaps of privacy metrics across three scenarios for synthetic and de-identified datasets (All Columns, *ACSIncome* dataset)

are represented by their means, and unseen categorical values are replaced with ‘*’. The pipeline is implemented using Scikit-learn. Utility is measured as predictive accuracy using *ML Efficacy* (MLE).

For *Patients*, utility is evaluated through an optimisation model following [6], implemented using Pyomo with the Gurobi solver. Instead of averaging generalised values as in [6], we apply a tailored domain transformation technique.

5 Quantitative Results

5.1 Privacy Evaluation and Comparison

We first evaluate the diversity of our metric suite, specifically by investigating the extent to which their results are statistically correlated. Then, we present the overall results expressed in term of privacy gains ($PG(M)$ with $M \in \{A, S\}$).

Privacy metric correlations. Figure 2 illustrates the pairwise correlations, specifically Spearman’s ρ , between the pairs of privacy metrics⁷. We observe strong correlations within families of similarity-based metrics (such as SR, NNA, NND) and within attack-based metrics (such as AIA-C and AIA-R) across all settings. A key distinction emerges between synthetic and de-identified data. For synthetic datasets, similarity-based metrics exhibit less or even negative correlation with attack-based metrics, suggesting that similarity to the original data does not directly translate into increased vulnerability to inference attacks. In contrast, for de-identified data, similarity-based metrics are consistently correlated with AIA, indicating that greater similarity to the original data is associated with higher attack success rates. This occurs because generalized attribute values can still serve as predictors of sensitive attributes in AIA.

This difference reflects the fundamentally distinct privacy mechanisms of the two approaches. De-identification operates directly on original records, such that residual similarity preserves information exploitable by inference attacks. As a result, similarity serves as a proxy for attack risk in de-identified datasets. Synthetic generation, by contrast, decouples record-level similarity from downstream privacy leakage, breaking the direct link between similarity-based estimators and attack effectiveness. Consequently, similarity-based metrics are less predictive of attack vulnerability for synthetic data.

⁷The conclusions drawn from the privacy metric correlation analysis are consistent across the *ACSIncome* and *Patient* datasets, as well as across the All-columns and QID attribute dimensions. Due to space limitations, we report results for *ACSIncome* only; the complete results are provided in the supplementary materials [1].

Table 3: Privacy gain comparison (values are printed as ‘mean±standard dev.’).

Method	ACSIIncome					
	PG _{NND}		PG _{DR}		PG _{AIA-R}	
	QID	ALL	QID	ALL	QID	ALL
CTGAN	1.23 ± 0.00	1.21 ± 0.00	0.98 ± 0.00	1.03 ± 0.00	1.46 ± 0.04	1.61 ± 0.01
DDPM	2.01 ± 0.01	1.93 ± 0.01	1.32 ± 0.00	1.31 ± 0.00	1.97 ± 0.03	2.23 ± 0.08
TVAE	1.49 ± 0.00	1.44 ± 0.00	1.30 ± 0.00	1.30 ± 0.00	1.30 ± 0.01	1.37 ± 0.06
k=5	2.39 ± 0.34	2.25 ± 0.32	1.23 ± 0.04	1.22 ± 0.04	1.29 ± 0.12	1.40 ± 0.12
k=10	2.49 ± 0.34	2.35 ± 0.32	1.24 ± 0.04	1.23 ± 0.03	1.33 ± 0.11	1.44 ± 0.12
k=20	2.57 ± 0.32	2.42 ± 0.30	1.25 ± 0.03	1.24 ± 0.03	1.36 ± 0.10	1.47 ± 0.11

Method	Patients					
	PG _{NND}		PG _{DR}		PG _{AIA-R}	
	QID	ALL	QID	ALL	QID	ALL
CTGAN	1.12 ± 0.00	1.12 ± 0.00	0.75 ± 0.01	0.91 ± 0.00	1.51 ± 0.01	1.81 ± 0.01
DDPM	1.67 ± 0.01	1.50 ± 0.01	1.24 ± 0.00	1.33 ± 0.00	2.65 ± 0.06	3.72 ± 0.18
TVAE	1.18 ± 0.00	1.12 ± 0.00	1.14 ± 0.00	1.28 ± 0.00	1.23 ± 0.01	1.45 ± 0.01
k=5	2.59 ± 0.58	1.77 ± 0.40	0.81 ± 0.14	0.86 ± 0.14	1.32 ± 0.13	1.75 ± 0.14
k=10	2.61 ± 0.56	1.77 ± 0.38	0.81 ± 0.13	0.86 ± 0.13	1.34 ± 0.11	1.78 ± 0.11
k=20	2.62 ± 0.53	1.78 ± 0.36	0.81 ± 0.13	0.86 ± 0.13	1.36 ± 0.10	1.80 ± 0.09

Table 4: Utility loss comparison.

Method	ACSIIncome		Patients	
	UL _{MLE}	CQE	RAU	CQE
CTGAN	0.35 ± 0.01	0.74 ± 0.00	1.00 ± 0.00	1.20 ± 0.06
DDPM	0.41 ± 0.02	1.53 ± 0.02	1.09 ± 0.03	5.61 ± 0.04
TVAE	0.29 ± 0.01	0.71 ± 0.01	1.01 ± 0.00	0.75 ± 0.01
k=5	0.25 ± 0.03	3.30 ± 1.43	1.01 ± 0.01	16.64 ± 10.13
k=10	0.26 ± 0.03	3.58 ± 1.40	1.01 ± 0.01	16.45 ± 10.17
k=20	0.26 ± 0.03	3.79 ± 1.35	1.01 ± 0.01	16.30 ± 10.22

Privacy gains. We select representative metrics from each privacy metric family⁸ and present their results. Specifically, we report PG_{NND} (nearest-neighbour distance) as the primary similarity-based metric, together with PG_{DR} (detection rate) as the primary model-based privacy metric, and PG_{AIA-R} (attribute inference attack, regression) as a representative attack-based metric. Table 3 presents the privacy gains under both QID-only and all-columns attribute dimensions.

De-identified datasets consistently achieve higher PG_{NND} than synthetic datasets across all scenarios. This outcome is expected, as de-identification techniques explicitly apply generalisation and suppression to QIDs, thereby increasing the distance between protected records and their nearest neighbours in the original data (this is also reflected in the $k = 20$ setting, which represents the most significant distortion). In contrast, synthetic data generation does not significantly disturb QIDs’ values, resulting in more moderate gains under similarity-based metrics.

For the *ACSIIncome* case, PG_{DR} are comparable for synthetic and de-identified data, with synthetic data exhibiting a slightly higher gain. This suggests that, in this setting, both approaches yield a similar level of distributional separation from the original data. In contrast, for the *Patient* case, synthetic data achieves substantially higher PG_{DR} than de-identified data. This can be attributed to the stronger cross-attribute correlations present in the *Patient* dataset, where generative models introduce more pronounced distributional changes than QID-based generalisation. As a result, the distribution of synthetic data is more skewed than that of the original and de-identified datasets.

Synthetic datasets generally yield higher attack-based privacy gains, as measured by PG_{AIA-R}, than de-identified datasets. This indicates that although de-identified data may exhibit more distortion according to similarity-based metrics; Residual attribute dependencies remain exploitable by inference attacks. Synthetic data generation, by decoupling records from the original dataset, effectively disrupts attribute-level inference and distribution-based signals.

Discussion. These results indicate that similarity-based privacy gains and attack-based privacy gains capture fundamentally different aspects of privacy. While stronger dissimilarity may indicate compliance with legislative privacy requirements (e.g., anonymity requirements under the GDPR), it does not necessarily imply stronger resistance to inference attacks. This distinction highlights the importance of reporting privacy metrics that align with the intended threat model and the actual privacy goals (i.e., protecting anonymity or sensitive values), rather than relying on aggregate scores across heterogeneous measures.

5.2 Utility Evaluation and Comparison

We first present the utility loss results, and then discuss the trade-offs between utility and privacy.

⁸The results of the complete metric suite can be found in the supporting materials [1].

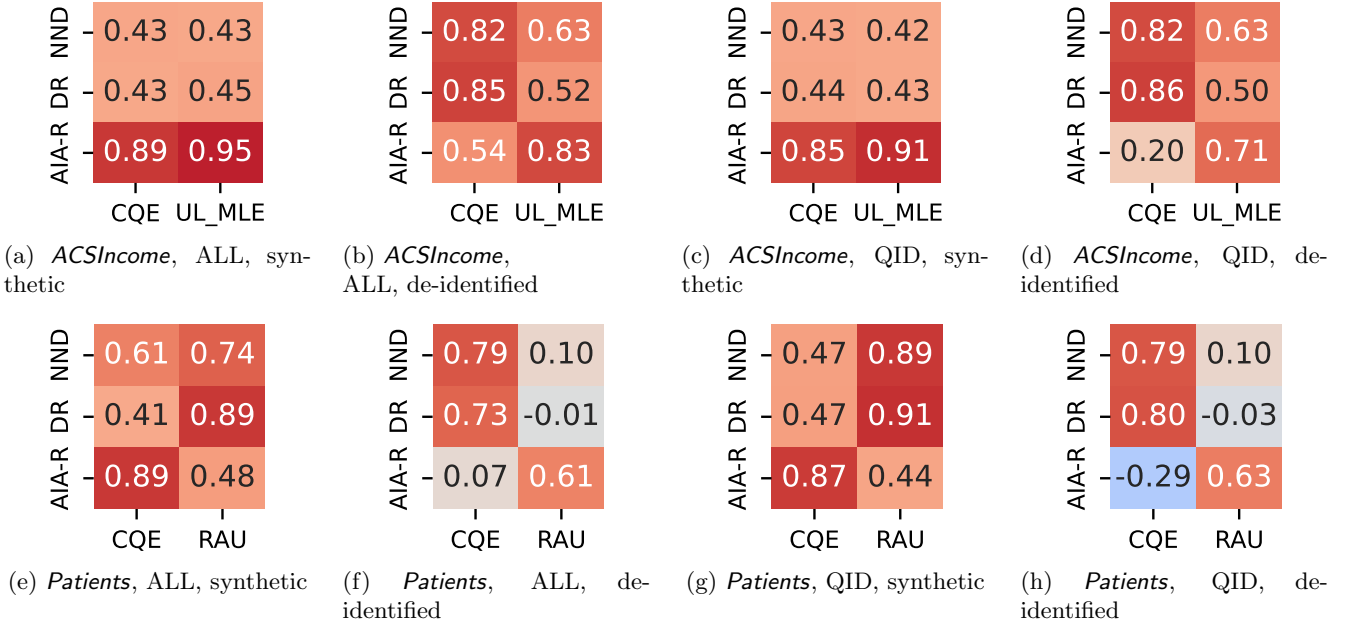


Figure 3: Privacy-utility cross-correlation matrices.

Utility Loss comparison. Table 4 compares the utility losses across the synthetic and de-identified datasets. RAU exhibits comparable values across the synthetic and de-identified datasets, whereas UL_{MLE} and CQE results diverge.

For the *ACSIncome* dataset, de-identified data consistently yields lower machine-learning (ML) loss than synthetic data. This result could be attributed to the fact that supervised ML models are designed to learn *generalizable patterns* rather than to memorise individual records⁹. De-identification techniques based on generalisation transform fine-grained attributes into coarser representations, thereby reducing noise and variance in the training data and facilitating the learning of robust decision boundaries.

In contrast, CQE reveals that query utility is substantially better preserved in synthetic data than in de-identified data across both datasets. This gap can be explained by generative models’ ability to capture multivariate joint distributions and attribute correlations. De-identification methods based on generalisation implicitly assume uniformity within equivalence classes and therefore discard fine-grained distributional information, leading to higher query distortion, especially for conjunctive or range-based queries.

Privacy-Utility Tradeoff. We construct cross-correlation matrices between privacy and utility metrics to analyse how privacy-utility trade-offs differ between de-identified and synthetic datasets for the *ACSIncome* and *Patients* cases. Figure 3 reports Spearman correlations between privacy metrics (where higher values indicate stronger privacy protection) and utility loss metrics (where higher values indicate lower utility). Positive correlations indicate privacy-utility trade-offs, with larger coefficients reflecting a tighter coupling between privacy gains and utility degradation.

For de-identified datasets, both the similarity-based metric NND and the model-based metric DR exhibit consistently strong correlations with count query error CQE across both datasets ($0.73 \leq \rho \leq 0.86$). In contrast, synthetic datasets show only moderate correlations ($0.41 \leq \rho \leq 0.61$). This indicates that, under de-identification, increases in record-level dissimilarity or distributional distortion are more directly coupled with degradation in query accuracy. The stronger coupling can be explained by the fact that generalisation-based de-identification directly alters attribute ranges and frequency structure, which are precisely the features exploited by count queries.

The privacy-utility trade-off for machine learning tasks is comparatively consistent across synthetic and de-identified datasets. In both cases, privacy metrics associated with resistance to attribute inference (e.g., AIA-R) exhibit stronger correlations with ML utility loss than similarity/model-based metrics such as NND or DR. This suggests that degrading attribute predictability, rather than merely increasing record-level dissimilarity, has a more pronounced impact on downstream predictive performance, regardless of whether privacy is achieved through synthesis or de-identification.

For resource allocation optimisation tasks, data synthesis and de-identification exhibit markedly different trade-off patterns. In synthetic datasets, optimisation utility loss correlates strongly with both NND and DR ($0.74 \leq \rho \leq 0.91$), indicating that increased value-level and distributional distortion substantially degrade optimisation performance. By contrast, correlations are weak or negligible for de-identified datasets. A likely explanation is that de-identified data remain grounded in real records, such that optimisation solutions derived from generalised but authentic locations are less sensitive to privacy-induced distortion than those derived from fully synthetic data.

Discussion. The utility results presented above indicate that the optimal PPDP strategy, from a utility perspective, strongly depends on the targeted use case. De-identified datasets outperform synthetic datasets in terms of *ML*

⁹This advantage, however, relies on the assumption that practitioners know the applied generalisation scheme. We note that this assumption does not introduce additional privacy risks, as such generalisation rules can typically be inferred from released data or prior deployments.

Utility, provided that suitable generalisation hierarchies can be defined; otherwise, synthetic data may be preferable, as it does not require additional, use-case-specific pre-processing or post-processing for machine learning tasks. In contrast, synthetic datasets consistently achieve lower *Count Query Error*, indicating better preservation of statistical query utility. For *Optimisation Utility*, both techniques can deliver comparable and satisfactory performance.

Across both datasets, the strength and nature of the observed privacy-utility trade-offs depend critically on the *family of privacy metrics* considered. Similarity-based metrics primarily capture distortion at the record level and are therefore most informative for de-identified data. In contrast, attack-based metrics capture inference resistance and are more indicative of utility degradation in synthetic datasets. Taken together, these findings demonstrate that conclusions about privacy-utility trade-offs are susceptible to the choice of privacy metrics.

6 Discussion

6.1 Threats to Validity

Our study is subject to several potential threats to validity:

- **Metric selection:** Bias may have been introduced by excluding some privacy metrics (cf, Table 1). Metric selection is based on three criteria: (i) applicability to both outcomes, (ii) diversity, and (iii) enabling fair comparison.
- **Exclusion of differential privacy:** Differential privacy, as a privacy definition, is often used in combination with PPDP techniques, i.e., de-identification and synthetic data generation. To study the *isolated effects*, we have excluded differential privacy mechanisms from our evaluation.
- **Sampling bias:** Because domain transformation involves sampling, individual trials may yield different results, potentially affecting reproducibility. However, sampling bias is mitigated by the large dataset size and by drawing attribute values from the set of possible offspring of each generalised value.
- **No optimisations:** We adopted a best-effort approach to generating synthetic datasets, primarily relying on default parameters. We acknowledge that tailored optimisations performed by experts may yield better results.
- **External validity:** While the use cases in our study (i.e., the pharmacy outlet optimisation problem and the ML income prediction case) are indicative of typical PPDP scenarios, we refrain from generalising the findings more broadly.

6.2 Other relevant aspects of PPDP

We further compare the two PPDP approaches along four dimensions: required effort, flexibility, maturity, and hardware requirements.

Required effort. Both PPDP techniques require substantial effort and expertise. De-identification requires domain knowledge and understanding of the intended use case, in which practitioners must identify QIDs, identifiers, and sensitive attributes, define generalisation hierarchies, and manually select an appropriate privacy level. Synthetic data generation requires knowledge of statistical and AI modelling, as practitioners must select suitable generative models, carefully define hyperparameters, and tune model parameters.

Flexibility. De-identification techniques offer limited flexibility: adding rows or columns requires re-evaluating privacy levels, reassessing potential quasi-identifiers, and updating generalisation hierarchies when new attribute values are introduced. In contrast, synthetic data generation is more flexible, as models can be updated incrementally and new records can be sampled on-demand.

Technological maturity. De-identification tools are mature and user-friendly. Established systems such as ARX [31], and sdcMicro [40] support a wide range of de-identification methods, with ARX being the most mature and comprehensive. In contrast, synthetic data generation is more recent and rapidly evolving. Actively maintained libraries such as Synthcity [32] and the Synthetic Data Vault Project [28] integrate data generators with evaluation metrics, but remain largely primarily developer-oriented: they lack the beginner-friendly graphical interfaces and documentation of current de-identification tools.

Hardware requirements. Throughout our experiments, we have recorded the memory usage and processing time statistics of the applied de-identification and synthetic data generation techniques, which can be consulted in supporting materials [1]. Data de-identification proved substantially more efficient, typically completing within seconds, whereas training of the generative models ranged from minutes to days, depending on the model and dataset. Memory usage of generative models also varied considerably across datasets. While de-identification algorithms are often executed over a single core, generative models can exploit multi-core CPUs and GPUs to improve performance, at the cost of substantially higher RAM and VRAM requirements.

7 Related Work

We first discuss related studies on the efficacy of PPDP methods, and then we compare our framework with other tools.

Quantitative evaluation of PPDP Methods. To the best of our knowledge, only two existing studies in the literature [36,37] directly compare PPDP techniques on structure and tabular datasets. The first [36] assesses whether the publication of synthetic data rather than original data reduces information leakage. The second [37] directly compares the privacy and utility of synthetic and de-identified datasets. Both studies adopt similar evaluation methodologies by simulating MIA and AIA on released datasets and reporting privacy gains relative to the original or de-identified data, alongside utility assessments based on ML performance and summary statistics. [36,37] find that synthetic data is not a superior alternative to de-identification due to highly stochastic and unpredictable privacy and utility outcomes.

In contrast to [37] who facilitated k -anonymity via record removal, we adopt a representative de-identification strategy based on generalisation, leading to a more realistic and methodologically sound comparison. Moreover, our evaluation explicitly distinguishes between privacy effects at QID level and at the full-dataset level. This dual-granularity analysis provides more comprehensive insights into how de-identification impacts privacy.

Open Source Privacy Evaluation Frameworks. Table 5 outlines the different open-source tools that support the privacy evaluation of PPDP techniques. As shown in the final column, our framework provides the most comprehensive and diverse privacy metric suite. These privacy metrics were selected based on their applicability and relevancy to both synthetic and de-identified datasets.

Table 5: Comparison of open source privacy evaluation tools.

	[28]	[32]	[36]	[37]	[11]	Our framework
Nearest Neighbor Distance	◆	◆				◆●
Close Value Ratio	◆	◆				◆●
Similarity Ratio		◆				◆●
Nearest Neighbor Accuracy						◆●
Data Likelihood	◆					◆●
Detection Rate						◆●
Structural Similarity						◆●
Attribute Inference Attack	◆	◆	◆	◆●	◆	◆●
Membership Inference Attack		◆	◆	◆●		◆●
Singling out					◆	
Record Linkage					◆	
Hitting Ratio	◆	◆				

◆ Evaluation of synthetic data ● Evaluation of de-identified data

8 Conclusions

Data de-identification and synthetic data generation are widely used yet fundamentally different PPDP techniques. Their respective privacy evaluations rely on disparate metrics and assumptions, which makes a direct comparison difficult. To address this gap, we introduce a unified, novel privacy evaluation framework that applies to both synthetic and de-identified data and integrates widely used privacy notions from both domains. Developing this framework involved re-conceptualising and adapting existing metrics to ensure fair and consistent applicability of both PPDP techniques.

We evaluate the framework on two complementary use cases: a location optimisation task using the *Patients* dataset, and a predictive learning task using the *ACSIncome* dataset. Our results show that de-identified data performs better under similarity-based privacy metrics, whereas synthetic data offers stronger protection against adversarial evaluations such as attribute inference (AIA), while both approaches achieve comparable utility.

Overall, neither synthetic data nor de-identified data is universally superior, and rather than drawing a definitive conclusion on the superiority of a PPDP technique, we emphasise the importance of comprehensive, multi-metric evaluation. Their suitability strongly depends on the intended use case and risk tolerance. This key insight further highlights the importance of our unified evaluation framework, which enables practitioners to assess privacy-utility trade-offs in a principled and reproducible manner and to make informed decisions about appropriate PPDP strategies in an application-aware context.

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Open Science Statement. The framework implementation, including the ported and adapted privacy and utility metrics and the *ACSIncome* dataset, is available in the supporting materials [1]. We exclude the *Patients* dataset because Kaggle retracted it during this research.

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This version of the contribution has been accepted for publication after peer review but is not the Version of Record and does not reflect post-acceptance improvements or corrections.

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